

CANOD Consumer Utility Audit

AUDIT REPORT

Comprehensive Evaluation of Articles, Tools, Technical Jargon, and Content Gaps for Everyday Consumers

Target Audience: Everyday Consumers (Homeowners, Remote Workers, Families) **Date:** September 2026 **Scope:** 11 Published Guides • 8 Interactive Tools • 20 Content Gaps

Executive Finding: CANOD's engineering foundation is exceptionally rigorous—its computational permutation engines, Canadian regulatory compliance references (CSA, ESA, Health Canada, CATSA), and technical accuracy are best-in-class. However, several existing articles and tools suffer from *cognitive friction and protocol-first framing* (e.g. demanding users identify "DisplayPort Alt Mode" or calculate RAID parity). CANOD must pivot from an "engineer's compliance manual" into an **"instant problem solver"** where technical terminology strictly follows simple, 30-second answers.

PART 1 — ARTICLES AUDIT (All 11 Published Guides)

ARTICLE TITLE & SLUG	MAIN SEARCH QUESTION ANSWERED	AUDIENCE	TECH	UTILITY	FAST?	COMMERCIAL	PRODUCT CATEGORIES	VERDICT
Why Is My Phone Charging Slowly? /why-is-my-phone-charging-slowly/	"Why is my phone taking forever to charge?"	Smartphone users	3	9.5	✔ Yes	High	25W–45W GaN wall chargers, braided USB-C cables	KEEP
Why Is My Wi-Fi Slow in One Room? /wifi-slow-in-one-room/	"Why does Wi-Fi crawl in the bedroom upstairs?"	Homeowners, renters	3	9	✔ Yes	Med-High	Wi-Fi 6 routers, mesh nodes, flat Ethernet cables	KEEP
What Size Power Bank Do I Need? /what-size-power-bank-do-i-need/	"What size portable battery should I buy for flights?"	Travelers, commuters	4	8	⚠ Mod	High	10,000 mAh pocket power banks, 65W laptop banks	KEEP
How Much Backup Storage Do I Need? /how-much-backup-storage/	"What size external SSD do I need to back up files?"	Mac & Windows owners	4	8	✔ Yes	High	1TB–2TB portable SSDs, desktop backup drives	KEEP
Do I Need a Router or Mesh System? /router-or-mesh-system/	"Should I buy a new router, extender, or mesh kit?"	Multi-floor owners	4	8	✔ Yes	High	Whole-home mesh kits (Deco, Eero), Wi-Fi 6 routers	KEEP
Why USB-C Cables Work Differently /why-usb-c-cables-work-differently/	"Why doesn't my cable charge my laptop or run monitor?"	Everyday tech users	5	8	✔ Yes	High	100W/240W charge cables, Thunderbolt 4 cables	KEEP
Choosing a Tech Organizer /choosing-a-tech-organizer/	"What is the best travel pouch for cables & chargers?"	Commuters, remote workers	2	7	✔ Yes	High	Cable pouches, travel desk organizers, cord wraps	KEEP
Power Bar or Surge Protector: Canadian Guide /power-bar-or-surge-protector-canada/	"Difference between power bar & surge protector?"	Home office & TV buyers	6	7	✔ Yes	High	Surge protectors with USB, heavy duty power bars	SIMPLIFY
Before Buying a Charger in Canada: Safety Checks /charger-safety-canada/	"How do I know if an online charger is safe in Canada?"	Replacement charger buyers	8	5	✗ No	High	Certified GaN wall chargers, multi-port desk chargers	SIMPLIFY
Seven Things Before Buying a USB-C Dock /seven-things-usb-c-dock/	"How to connect two monitors to laptop with 1 cable?"	Desk workers, multi-monitor	9	4	✗ No	High	Dual-monitor docks, Thunderbolt 4 hubs, DisplayLink	REWRITE
Choosing a Home NAS: Storage & Backups /choosing-a-home-nas/	"Should I get a NAS or just an external hard drive?"	Enthusiasts (alienates users)	9	3	✗ No	Medium	2-bay NAS enclosures, 3.5" NAS drives, external drives	REWRITE

Flagged Acronyms & Engineering Jargon

DP Alt Mode MST USB4/TB4 EPR (240W) DisplayLink DSC
RAID 0/1 Diskless 1GbE ESA Rule 2-024 CATSA

Rule: Never lead with these terms in headlines or intro paragraphs. A consumer must be told *"Charging cords do not carry video"* before any mention of DP Alt Mode.

Title Optimization for Natural Search

CURRENT EDITORIAL TITLE	RECOMMENDED NATURAL SEARCH TITLE
Seven Things to Check Before Buying a USB-C Dock	How to Connect Two Monitors to a Laptop (Mac & PC)
Before Buying a Charger in Canada: Seven Safety Checks	Is This Charger Safe to Buy in Canada? (Look for These 3 Logos)
Power Bar or Surge Protector: What Should Canadians Check?	Power Bar vs Surge Protector: Which Do You Actually Need?
Choosing a Home NAS: Storage, Backups & Compatibility	External Hard Drive, Cloud, or NAS: Which Do You Need?

PART 2 — TOOLS AUDIT (Calculators & Interactive Checkers)

Inspection of all 8 active and planned tools in `lib/tools-data.ts`, including code-level inspection of `dock-checker.ts` and `safety-checker.ts`.

Tool Name & Slug	Status	Consumer Problem Solved	Inputs Required	Outputs Provided	Tech	Utility	Verdict
USB-C Dock & Dual Monitor Checker <code>/tools/usb-c-dock-checker/</code>	LIVE	"Will this dock run two extended monitors and charge my laptop?"	Device, OS, Port type (DP Alt Mode, USB4), Monitors, Res, Watts	Compatibility report, Mac MST warning, DisplayLink alert, power check	8	6	SIMPLIFY
Canadian Electrical Safety Checklist <code>/tools/canadian-electrical-safety-checklist/</code>	LIVE	"Is this unbranded charger or power bar a fire hazard?"	Product type, Mains power, Approval marks, Specs, Seller, Recall	Confirmed/Verify/Concern status, safety warnings, registry links	5	5	SIMPLIFY
Dual Monitor Hub & Bandwidth Calculator Active via Dock Suite	LIVE	"Can my laptop run two 4K screens at 60 Hz without lagging?"	Processor (Mac/PC), Target resolution, Refresh rate (60/120 Hz)	Display lane allocation assessment, Mac warning, port sequence	7	6	MERGE
Charger Wattage Calculator <code>/tools/#charger-wattage-calculator</code>	SOON	"What wattage charger do I need for my phone & laptop together?"	Laptop model, Phone model, Tablet, Simultaneous devices, Travel/Desk	Recommended total wattage (65W/100W GaN), port splitting preview	4	10	BUILD NEXT
USB-C Cable Capability Checker <code>/tools/#usbc-cable-checker</code>	SOON	"Will the cable in my drawer charge my laptop or run a monitor?"	Intended task (power, data, monitor), Cable tip markings, Length	Verdict: Full power / Slow charging / No video + shopping guidance	4	9	BUILD NEXT
Wi-Fi Coverage & Node Calculator <code>/tools/#wifi-coverage-calculator</code>	SOON	"Do I need a single router, an extender, or a 2/3-pack mesh kit?"	Home size (sq ft), Floors/basement, Wall material, Modem location	Verdict: Router vs 2-pack Mesh, node placement map, wiring tips	5	9.5	BUILD NEXT
Backup Storage Capacity Calculator <code>/tools/#backup-storage-calculator</code>	SOON	"What size external SSD should I buy so it doesn't fill up in 6 mo?"	Computer drive size, Annual photo/video growth, Desired history years	Recommended drive size (1TB/2TB/4TB), 3-2-1 backup checklist	3	8	KEEP
NAS Capacity & RAID Calculator <code>/tools/#nas-capacity-calculator</code>	SOON	"How much usable space do I get after RAID mirroring?"	Bays count (2/4), Drive capacity (4TB/8TB), Mirroring safety	Usable space vs parity drive space vs filesystem formatting loss	6	3	REMOVE

1. USB-C Dock Compatibility Engine (`lib/dock-checker.ts`)
Assessment: 25,200 permutations. Technically brilliant: accurately flags Mac MST mirror limitations and DisplayLink permissions. **Flaw:** Asks users to choose "DP Alt Mode" vs "USB4". *Must replace with a simple laptop brand/model picker.*

2. Canadian Electrical Safety Engine (`lib/safety-checker.ts`)
Assessment: 183,708 permutations. Safely enforces dependency checks. **Flaw:** 10 self-reported form fields asking users to read tiny embossed text and search federal databases. *Must rebuild into a 3-step visual "Safe or Scam" screener.*

PART 3 — CONTENT GAPS: 20 Everyday Consumer Problems (Part I)

High-intent consumer search problems CANOD should prioritize over engineering topics (Items 1 to 10).

1. Laptop Won't Charge Over USB-C

Search: "laptop plugged in not charging usb-c"

User tries charging a 65W laptop with a 15W phone charger or unrated cable; laptop shows warning icon or slowly drains under load.

Monetize: 65W–100W GaN chargers, 100W/240W USB-C cables

2. Laptop Won't Detect External Monitor ("No Signal")

Search: "laptop not detecting monitor hdmi usb-c"

User plugs HDMI or USB-C into a monitor and gets a black screen. Diagnosing cable mismatch, input source, and display driver settings.

Monetize: Certified USB-C to HDMI/DP cables, HDMI 2.1 cords

3. How to Connect Two Monitors to a Laptop (Mac vs PC)

Search: "how to connect two monitors to macbook air"

Explains why standard docks mirror both screens on Apple Silicon Macs (MST limitation) and shows the exact dock/adaptor to get two extended screens.

Monetize: Dual-display docks, DisplayLink adapters

4. Which Charger Do I Need for My Phone and Laptop?

Search: "one charger for phone and laptop"

How to replace 3 bulky power bricks with one compact travel charger. Explains simultaneous wattage splitting in simple English.

Monetize: Multi-port 65W–100W GaN travel plugs

5. Which HDMI Cable Do I Actually Need for My TV / Monitor?

Search: "difference between hdmi cables 4k 120hz"

Saves consumers from buying \$80 "gold-plated 8K" cables when a \$14 certified Premium or Ultra High Speed HDMI cable does the job.

Monetize: Certified 18Gbps (HDMI 2.0) and 48Gbps (HDMI 2.1) cables

6. External SSD vs Portable Hard Drive: Which Should You Buy?

Search: "external ssd vs hard drive for backup"

Explains why SSDs cost more but survive drops, transfer photos 5x faster, and don't fail when bumped during a vacation backup.

Monetize: Rugged portable SSDs (Samsung T7, Crucial X9)

7. How to Back Up Phone Photos Without Paying for Cloud Storage

Search: "backup iphone photos to external drive no icloud"

How to move thousands of photos from an iPhone or Android to a portable SSD or flash drive to escape recurring \$4–\$15 monthly cloud fees.

Monetize: Dual USB-C flash drives, portable backup SSDs

8. USB-C Cable Won't Display Video on My Monitor

Search: "usb c cable not working with monitor"

Demystifies why a charging cable that charges a phone fine gives a black screen on a monitor (charging vs high-speed video wiring).

Monetize: Certified USB-C Video and Thunderbolt 4 cables

9. Travel Chargers for Canadians: Adapters vs Converters

Search: "canadian plug adapter for europe voltage"

Explains plug shape vs voltage conversion so travelers don't melt hair curlers or fry phone chargers in Europe, Asia, or the UK.

Monetize: Universal all-in-one travel adapters with USB-C

10. Cable Organization for a Clean Desk: The Beginner's Setup

Search: "how to hide cables under desk home office"

Practical step-by-step guide to routing power bricks, monitor cables, and laptop cords for an uncluttered, single-cable workspace.

Monetize: Under-desk wire trays, velcro ties, cable boxes

PART 3 (Cont'd) — CONTENT GAPS: 20 Everyday Consumer Problems (Part II)

High-intent consumer search problems CANOD should prioritize over engineering topics (Items 11 to 20).

11. Why Does My Wi-Fi Drop or Freeze During Zoom / Teams Calls?

Search: "video calls lag but internet speed test is fast"

Explains bufferbloat, jitter, and microwave/Bluetooth interference when download speeds are fine but live audio and video stutter.

Monetize: Wi-Fi 6 routers with QoS, flat Ethernet cables

12. Why Is My Phone Getting Unusually Hot While Charging?

Search: "phone gets hot while charging fast"

When heat is normal during fast charging vs when thick cases, direct sunlight, or failing batteries cause thermal throttling.

Monetize: Ventilated wireless charging stands, optimized chargers

13. Why Is My Bluetooth Mouse or Keyboard Stuttering / Lagging?

Search: "wireless mouse cursor lagging stuttering"

Reveals how USB 3.0 ports radiate RF interference that jams 2.4 GHz wireless mouse dongles, and how a short extension fixes it.

Monetize: Short USB extender cords, reliable Bluetooth mice

14. How to Connect Old USB Flash Drives to a New USB-C Laptop

Search: "how to plug normal usb into macbook"

A plain-English guide to cheap 2-pack USB-A to USB-C adapters and small travel hubs for users whose new laptop has only oval ports.

Monetize: Compact USB-C to USB-A adapters, mini hubs

15. Why Won't My Power Bank Charge My Laptop?

Search: "can power bank charge my macbook laptop"

Explains why a portable battery that charges an iPhone easily does nothing when plugged into a laptop (requires 45W–65W USB Power Delivery).

Monetize: 65W+ laptop-capable power banks (Anker 737, Baseus)

16. Why Is My External Hard Drive "Read-Only" on Mac?

Search: "cannot copy files to external hard drive mac"

Explains Windows NTFS vs Mac APFS vs cross-platform exFAT so users can share drives between Windows and Mac without losing files.

Monetize: ExFAT pre-formatted cross-platform portable drives

17. How to Clean a Dirty USB-C Charging Port Safely

Search: "charging cable loose falls out of phone"

How compressed pocket lint prevents cables from clicking in, and how to safely clean the port with non-conductive tools without damaging pins.

Monetize: Anti-static cleaning tools, port dust plugs

18. Why Is Wireless Charging So Slow (or Stopping at 80%)?

Search: "magsafe wireless charging slow iphone"

Explains thermal throttling, coil misalignment, thick cases, and why wireless charging is inherently slower and hotter than a cable.

Monetize: Certified MagSafe / Qi2 15W stands with active cooling

19. Why Won't My Smart Bulb or Robot Vacuum Connect to Wi-Fi?

Search: "smart plug won't connect to 5ghz wifi mesh router"

The 2.4 GHz vs 5 GHz single-SSID mesh trap that breaks smart home pairing, and how to temporarily split bands or use an IoT guest network.

Monetize: Dual-band routers with dedicated IoT network support

20. Is It Safe to Leave Laptops and Phones Plugged in Overnight?

Search: "does leaving laptop plugged in damage battery"

Debunks charging myths, explains modern charge-controller circuits, and shows how 80% charge limits help long-term battery health.

Monetize: Certified Canadian power bars with thermal cutoff

PART 4 — FINAL STRATEGIC RECOMMENDATIONS

A. Best 5 Existing Articles

- 1. **Why Is My Phone Charging Slowly?** Flagship article. 15-minute diagnostic test, room-temp checks, lint inspection. Absolute benchmark for the site.
- 2. **Why Is My Wi-Fi Slow in One Room?** Isolates single-device issues from building layout and router placement before pushing hardware.
- 3. **What Size Power Bank Do I Need?** Solves the 10,000 mAh vs real watt-hours dilemma and clarifies Canadian airport 100 Wh rules.
- 4. **Why USB-C Cables Work Differently:** Demystifies identical cables using the relatable Apple charge cable vs Thunderbolt comparison.
- 5. **Do I Need a Router or Mesh System?** Saves families hundreds of dollars by helping them avoid unnecessary 3-pack mesh systems.

B. Worst 5 Existing Articles

- 1. **Choosing a Home NAS:** Reads like an enterprise IT guide (RAID parity, diskless bays, 1GbE ceilings). Completely misaligned with ordinary consumers.
- 2. **Seven Things Before Buying a USB-C Dock:** Paralyzes users with DP Alt Mode, MST, and DisplayLink permissions before solving their monitor setup.
- 3. **Before Buying a Charger in Canada:** Valuable safety points, but reads like an ESA compliance brief citing Ontario rules.
- 4. **Power Bar or Surge Protector:** Gets bogged down in joule clamping voltages and provincial bulletins rather than simple buying rules.
- 5. **Choosing a Tech Organizer:** Good advice, but lowest consumer problem urgency and thinnest search volume on the site.

C. Best Existing Tool

USB-C Dock Compatibility Checker (/tools/usb-c-dock-checker/)

Its 25,200-permutation client-side engine in lib/dock-checker.ts is technically flawless. It accurately diagnoses Apple Silicon Mac dual-monitor mirroring and DisplayLink screen-recording permissions. *Once protocol dropdowns are replaced by a simple laptop selector, it will be CANOD's most valuable tool.*

D. Tools to Remove or Rebuild

- REMOVE: NAS Capacity Calculator:** Everyday consumers do not buy 4-bay NAS enclosures or configure RAID. Synology already provides a free RAID calculator. Replace with a *Photo & Cloud Storage Cost Calculator*.
- REBUILD: Canadian Electrical Safety Checklist:** Rebuild the 10-question self-report form into a 3-step visual *"Safe or Scam Screener"* with photos of genuine vs fake CSA marks.

E. The FIRST 10 Articles CANOD Should Improve / Create

PRIORITY	ACTION	WORKING TITLE	CORE OBJECTIVE
#1	REWRITE	How to Connect Two Monitors to a Laptop (Mac & PC)	Solve the dual extended display puzzle without confusing users with MST jargon.
#2	SIMPLIFY	Is This Charger Safe to Buy in Canada? (Look for These 3 Logos)	Visual guide showing genuine CSA/cUL logos vs fake CE stamps on Amazon/Temu.
#3	SIMPLIFY	Power Bar vs Surge Protector: Which Do You Actually Need?	Distinguish outlet expansion from surge suppression; keep the space heater warning prominent.
#4	CREATE	My Laptop Won't Charge Over USB-C: Quick Fixes	Diagnose underpowered phone bricks, loose lint in ports, and bad cables in 5 minutes.
#5	CREATE	Which Charger Do I Need for My Phone and Laptop?	The definitive guide to buying one multi-port GaN travel plug for all devices.
#6	CREATE	Laptop Won't Detect External Monitor: How to Fix "No Signal"	Step-by-step diagnostic workflow for HDMI, DisplayPort, and USB-C display blackouts.
#7	CREATE	Which HDMI Cable Do I Actually Need for My TV or Monitor?	Demystify HDMI 2.0 vs 2.1 without getting tricked by \$80 overpriced retailer cords.
#8	CREATE	External SSD vs Portable Hard Drive: Which Should You Buy?	Speed, durability, and reliability compared for backing up photos and computers.
#9	CREATE	How to Back Up Photos from Your Phone Without Monthly Cloud Fees	Save hundreds of dollars by moving photos directly to a dual-drive or portable SSD.
#10	CREATE	Travel Chargers for Canadians: Adapters vs Converters Explained	Prevent fried curling irons and dead laptops when traveling to Europe, UK, and Asia.

F. The FIRST 3 Tools CANOD Should Perfect

TOOL 1: BUILD NEXT

Charger Wattage Calculator

Calculates total watts and simultaneous port splitting (e.g. 65W laptop + 30W phone = 100W GaN). Solves the universal single-charger shopping dilemma. Highest affiliate monetization potential.

TOOL 2: SIMPLIFY EXISTING

USB-C Dock Compatibility Checker

Keep the brilliant 25,200 permutation engine, but remove protocol dropdowns ("DP Alt Mode"). Replace with a simple laptop brand/model picker. Solves the #1 desk setup headache for Mac & Windows.

TOOL 3: BUILD NEXT

Wi-Fi Coverage & Placement Calculator

Takes home square footage, floor count, and wall materials to deliver a clear verdict: "Single router vs 2-pack mesh", with interactive node placement advice that eliminates dead zones.